BLOG

Design Your Dream PC with CORSAIR

Building your own PC has become more popular than ever. Years ago, it seemed like a challenging task reserved only for tech enthusiasts. Today, thanks to advancements like the CORSAIR iCUE LINK Ecosystem, tasks such as cable management and lighting control can be done in minutes rather than hours, making the process simpler for everyone to design the PC of their dreams.



Getting started on your PC can be an empowering and a rewarding experience. Selecting each component brings you closer to a setup uniquely your own. Before beginning your build, clearly define what you want your PC to achieve whether it's purely for gaming, professional workloads, casual browsing, or high-quality streaming and explore how our CORSAIR iCUE LINK Ecosystem can help you reach them to design a PC that YOU have control over.



Why build your own PC?

Before diving into the hardware components that make up your PC, let's briefly discuss the benefits of building your own system.

Personalization: People build their own PCs because they crave control and personalization. It's more than just owning a computer it's about bringing your unique vision to life. Building your PC means creating something tailored precisely to your needs, designed with your own style, packed with exactly the features you want and none that you don't. Instead of settling for something off-the-shelf, you invest your time, effort, and research because nothing beats the satisfaction of crafting a system that's truly yours.

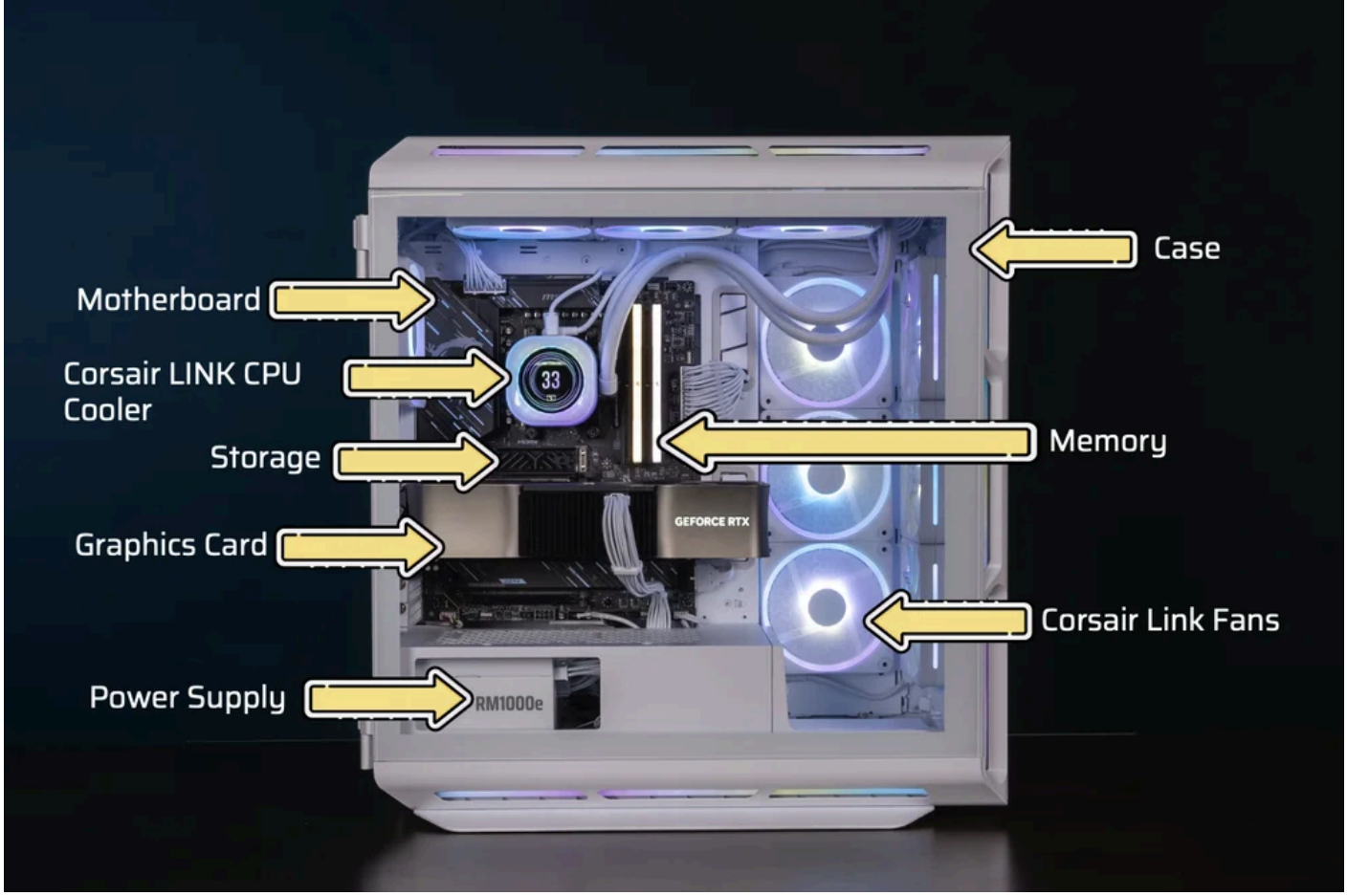
Upgradeability: As technology evolves or your needs change, you have the power to conveniently upgrade individual components, whether it's adding faster RAM, expanding your storage, or installing the latest graphics card. This freedom to swap parts effortlessly doesn't just keep your system ahead of the curve; it significantly extends its lifespan, making your custom build a lasting investment.

Experience: Not only will you gain valuable insight into how computers function internally, but you'll also find it much easier to troubleshoot issues in the future. The process itself serves as an excellent learning opportunity, and once you've mastered it, you can confidently share your expertise with friends and family. Consider building your own PC as an exciting project that encourages personal growth and deepens your understanding of technology.



The Layout

Here's a quick reference for a standard PC build. Each component is labeled for easy identification.



What components are inside a PC?

Every component in a PC has a specific role, and understanding how each one functions is essential. These components work together to ensure your PC operates as a unified system. Explore how our products can help you choose the right components for your build.

CPU

The CPU, or Central Processing Unit, is often called the “brain” of your PC (though not literally!). It handles all the instructions and commands needed to get tasks done. As one of your computer’s core components, the CPU has a huge impact on your system’s overall performance.

There are two well-known CPU manufacturers for general consumers: Intel and AMD.

The latest Intel® Core™ Ultra Processors and AMD Ryzen™ 9950X3D and 9900X3D CPUs demand powerful cooling solutions. Our high-performance AIO (All-in-One) liquid coolers, like our iCUE LINK TITAN RX, deliver exceptional cooling efficiency to manage the intensive workloads of these advanced processors.





GPU

A Graphics Processing Unit (GPU) also known as a Video Card or Graphics Card, handles the heavy lifting for demanding visual tasks, especially in gaming. Your GPU plays a big role in your PC's performance and overall gaming experience, impacting frame rates, resolution, visual effects, and more. It's an essential component for immersive gaming and professional workstation tasks.

Just like with CPUs, there are two major GPU makers - AMD and NVIDIA.

The latest NVIDIA GeForce RTX™ 50 series and AMD Radeon™ graphics 9000 series cards deliver unprecedented performance but demand substantial, stable power to operate at their best.

Our premium power supply units are engineered specifically for these rigorous power requirements, providing reliable, clean, and efficient energy delivery. With industry-leading certifications like Cybenetics and robust features including fully modular cabling, low ripple voltage, and exceptional load regulation, CORSAIR PSUs ensure your GPU operates efficiently, consistently, and at peak capability, no matter how intense the workload.





PC Case

The PC Case, also known as Chassis, Enclosure, PC Tower, or Cabinet, is where all your PC hardware is installed. It's the most visible component of your build, so if you want to showcase your build in the best way possible, you should go for a case that has good-looking aesthetics and is capable of accommodating different components like our iCUE Link Fans and Accessories . The CORSAIR FRAME 4000D and 7000 series are great examples of such cases. Our newest cases fully support reverse motherboard connectors including ASUS's BTF and MSI's Project Zero series.

For a few more lessons about PC cases, check our PC Cases 101, 102, and 103.

There are currently three common PC case sizes:

- Full-Tower
- Mid-Tower
- Small Form Factor



7000D AIRFLOW
Full-Tower



FRAME 4000D
Mid-Tower



2000D AIRFLOW
Small Form Factor

Depending on how much space you have allocated for your PC, choose the one that fits your setup. We offer a plethora of PC cases for all types of builds, ranging from Small Form Factor all the way up to Full Tower sizes.

Motherboard

The motherboard serves as your computer's central hub, connecting all its components together in one place. It lets your CPU, memory, and GPU communicate seamlessly, working together as one efficient team.

Just like PC cases, motherboards have different form factors or sizes. The four most common motherboards currently in the market are:

- EATX (Extended ATX)
- ATX (Standard)
- Micro-ATX
- Mini-ATX

Reverse connector motherboards, like ASUS BTF and MSI Project Zero, relocate power and peripheral connectors to the back for improved cable management, airflow, and a cleaner aesthetic. To support this innovation, CORSAIR's latest cases, including the 2500 and 6500 Series, are designed for full compatibility, featuring optimized cutouts and cable routing for seamless integration. Whether building with an ATX or micro-ATX reverse connector motherboard, CORSAIR cases ensure a streamlined, clutter-free setup without compromising performance or accessibility.

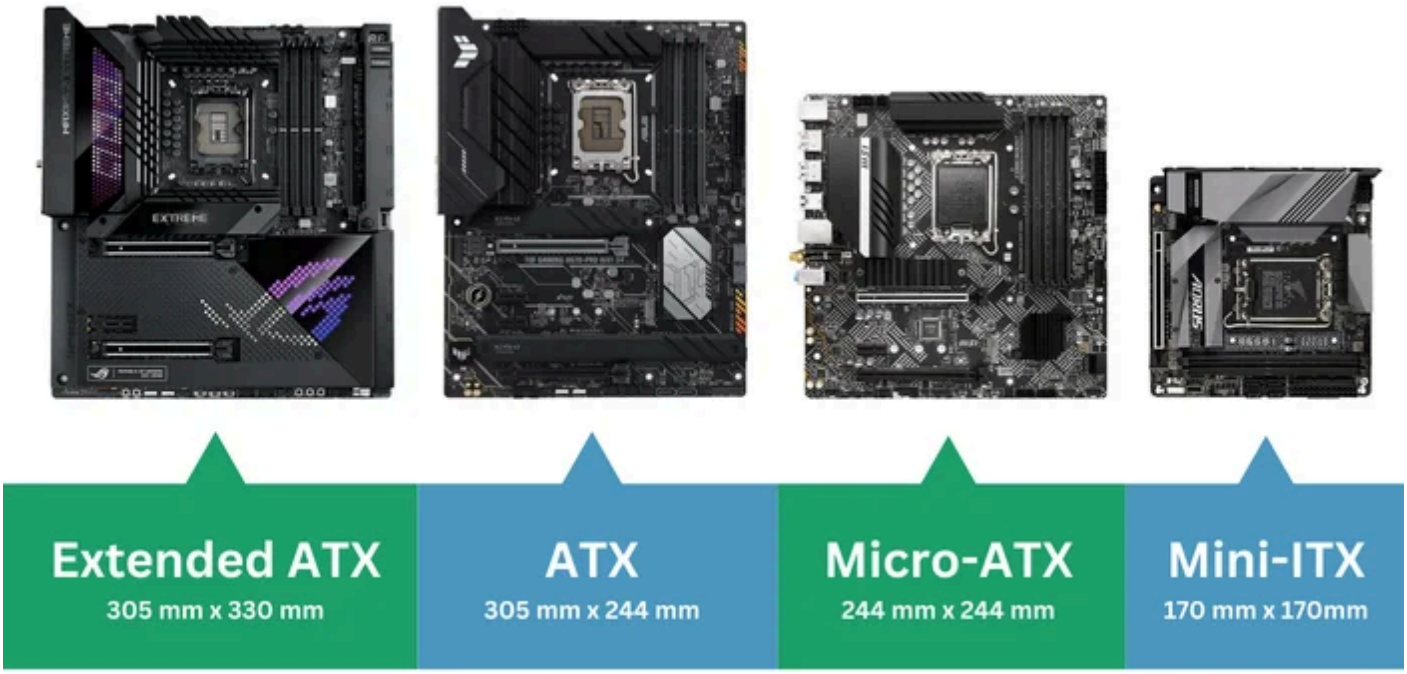


Image from StoneRefurb

Memory / DRAM

DRAM acts like your computer's desk space, giving applications fast temporary storage so you can handle more tasks at once. DDR5, the newest SDRAM generation, ramps up bandwidth, capacity, and power efficiency over DDR4. Perfect for gaming, content creation, and heavy-duty workloads, choosing the right amount of memory has never been simpler.

Founded as a pioneer in high-performance memory, CORSAIR's DDR5 modules feature rigorously screened high-frequency chips and onboard voltage regulation for precise overclocking. With support for Intel XMP 3.0 and AMD EXPO, they deliver optimal performance across all platforms.

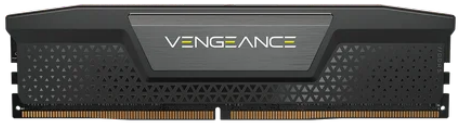
Explore our wide range of DRAM solutions, from the flagship DOMINATOR TITANIUM series to our sleek, low-profile VENGEANCE lineup.



DOMINATOR TITANIUM RGB DDR5



VENGEANCE RGB DDR5



VENGEANCE DDR5

CPU Cooler

A CPU cooler's main job is to keep your processor cool by dissipating heat. There are two types of CPU coolers: air cooling and liquid cooling. Right now, All-In-One (AIO) liquid cooling is the industry standard you'll find in most consumer PC builds.

Just like with our DRAM products, we've been at the forefront of innovation in AIO liquid CPU cooling for years and we still are. A perfect example is our iCUE LINK TITAN AIO Liquid CPU Coolers, a series designed specifically to integrate seamlessly with the CORSAIR iCUE LINK ecosystem.



Storage

PC storage devices, or drives, are where your computer stores digital data. They come in different capacities, most common are in gigabytes (GB) and terabytes (TB), allowing you to save everything from images and videos to games and software. Think of storage like a file cabinet in an office it keeps your important documents safe and accessible whenever you need them.

PC storage devices have evolved dramatically over recent years, especially with the arrival of NVMe M.2 SSDs.

You'll commonly find these types of drives on the market:

2.5-inch Drives: Compact and efficient, 2.5-inch drives are commonly used in laptops and desktops, balancing performance and storage capacity in a smaller form factor. They are ideal for systems where space is limited, offering reliability with decent speeds

3.5-inch Drives: Designed for desktops and servers, 3.5-inch drives provide higher storage capacities making them great for bulk storage and performance-heavy applications. They are the go-to choice for users needing maximum capacity at an affordable price.

NVMe M.2 drives: These drives leverage PCIe lanes from the Motherboard to deliver fast data transfer speeds, drastically reducing load times for games and applications. Their compact, cable-free design makes them perfect for modern desktops and laptops, maximizing both performance and efficiency in a minimal footprint.

CORSAIR has been there every step of the way, adapting as storage form factors changed, and continuously providing users with a wide range of high-performance storage options. Our NVMe drives deliver top-tier reliability and high-performance speeds, making them some of the best in the industry.



PCIe 5.0 (Gen 5) x4 NVMe M.2 SSD



SATA SSD 2.5 inch



3.5" Hard Drive from Seagate

Power Supply

The Power Supply Unit (PSU) might not be the flashiest part of your PC, but it's definitely one of the most important. The PSU converts alternating current (AC) from your wall outlet into stable direct current (DC), which your computer's components rely on. Its main job is to deliver clean, reliable power, keeping all your PC parts running safely and smoothly think of it as the heart pumping steady, consistent energy throughout your system.

By choosing CORSAIR's PSUs, you can power NVIDIA's and AMD's latest GPUs with confidence, ensuring stable performance and reliable wattage delivery for even the most demanding tasks.

There are three types of PSU cable configurations

- Modular PSUs
  - Fully Modular
  - Semi-Modular
- Non-Modular PSUs



Fully Modular All cables are detachable



Semi-Modular24pin & 8pin EPS cables are fixed; the rest are detachable



Non-Modular All cables are fixed

Fans

Fans are essential components that help keep your PC running smoothly. Since heat is the number-one enemy of hardware, poor cooling can quickly lead to premature component failure. Fans help by pulling cool air into your PC case and pushing warm air out, preventing overheating. Think of them as your PC's lungs, constantly breathing fresh air in and exhaling hot air out.

Not only do fans help make sure all your components are running at optimal performance, but PC fans can also be components that can improve a PC's overall aesthetics, especially now that they are capable of housing RGB lighting that can be customized to match your build's color scheme.

CORSAIR offers one of the widest range of PC case fans in the industry, from high-performance, such as the iCUE LINK RX120 RGB Fans, to smart and sophisticated designs, like the iCUE LINK QX RGB fans.

iCUE LINK fans are a key component in a PC that drastically changes how we build computers today. Before iCUE LINK fans, a standard fan required two cables to connect to a hub to function properly—this is not the case with iCUE LINK fans. You only need one type of cable to power all the fans in a PC build. This means there will be only a single cable for any fans positioned in a single row, creating the cleanest arrangement of fans to date. This cable connects back to the System Hub or other iCUE LINK compatible products (such as additional fans or an AIO cooler).





Accessories

CORSAIR iCUE LINK CapSwap VRM Fan Module

The CORSAIR iCUE LINK CapSwap VRM Fan Module is a simple yet effective upgrade for iCUE LINK AIO coolers, replacing the standard pump cap with a 70mm fan designed to enhance airflow over key motherboard components like VRMs, M.2 SSDs, and memory modules. With speeds of up to 3,000 RPM, it helps reduce thermal buildup, improving overall system stability especially for overclocked setups. The tool-free CapSwap design allows for quick and hassle-free installation, while the Magnetic Dome bearing ensures smooth and quiet operation. Engineered for seamless integration with iCUE LINK AIO coolers, this module provides an extra layer of cooling where it matters most.



iCUE LINK Aurora RGB Light Strips

The iCUE LINK Aurora RGB Light Strips bring dynamic RGB effects that sync seamlessly with your whole setup. Each strip features 40 individually addressable RGB LEDs, perfectly illuminating your case interior. With iCUE LINK, setup becomes even easier.

The iCUE LINK Aurora strips comes in two sizes:

350mm kit

430mm kit



Other Options

If you're planning to build your own PC, the CORSAIR PC Builder is the perfect tool to guide you in choosing the right components. Whether you're concerned about hardware compatibility or want to ensure you're picking the ideal CORSAIR iCUE LINK components, the PC Builder makes the selection process effortless. Use it to confidently create a clear game plan and make sure your build includes everything you need.

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